

Fuzzy Logic Based Condition Monitoring of a 3-Phase Induction Motor

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Abstract— Induction motors are widely used in transportation, mining, petrochemical, manufacturing and in almost every other field dealing with electrical power. These motors are simple, efficient, highly robust and rugged thus offering a very high degree of reliability. But like any other machine, they are vulnerable to faults, which if left unmonitored and detected, might lead to catastrophic failure of the machine in the long run. Early detection and diagnosis of faults that may occur are desirable to ensure that, operational effectiveness of an induction motor is improved. In this work, faults detection and classification using fuzzy logic controller is presented. A series of simulations using the models of 3-phase squirrel cage induction motor was performed using MATLAB/Simulink under different faults conditions, such as, break in stator windings, turn-to-turn short-circuit, 3-phase to ground and unbalance in input voltage. Models were designed on the basis of characteristics and parameters of a real motor. The results that were obtained showed that the fuzzy logic controller was able to detect the various faults of the 3-phase induction motor with high accuracy.

Keywords— Induction Motor, Fuzzy Logic Controller, Fault detection, break-in-stator winding fault, turn-to-turn short-circuit

I. INTRODUCTION

Induction motor is complex electro-mechanical device utilised for the conversion of power from electrical to mechanical form. Being a very prominent part of modern industry, it plays a vital role in major applications like pumping systems, elevating systems, crushers, and many more industrial segments. Depending on the rotor winding, induction motor is classified into two groups; squirrel-cage type and wound-rotor type. The speed of the induction motor can be controlled by varying synchronous speed and slip [1].

Although this motor is very reliable, it is prone to different modes of faults. These faults include; stator winding faults, broken rotor bars, eccentricity and bearing failures [2]. These faults may be inherent to the machine itself or due to operating conditions. If a fault persists, it may lead to plant failure, thus increased in production losses.

Several fault identification methods have been applied to detect machine faults at different stages by using different machine variables, such as current, voltage, speed, temperature and vibrations. A major difficulty of these approaches is the lack of well processing of input parameters. This project work

applied fuzzy logic controller, to monitor induction motor conditions, based on the amplitude features of the stator currents. In fact, fuzzy logic is based on human thinking process and natural language, enabling decisions to be made based on vague information [3].

II. INDUCTION MOTOR FAULTS

An induction motor is viewed as the heartbeat of the modern industry due to its extensive applications, low maintenance cost, easy control as well as high reliability. Induction motors are undoubtedly reliable but the possibility of induction motor failure cannot be avoided [2]. These failure conditions are taking place because of its component failure and they include stator faults, rotor faults, bearing failures and other external faults. The percentage failure components of induction motor are as shown in Fig. 1.

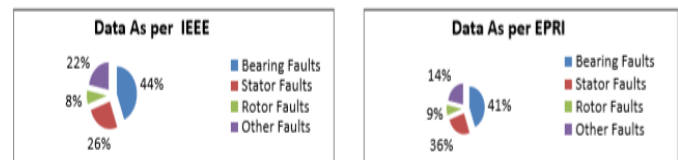


Fig. 1 Percentage component of induction motor failure per IEEE and EPRI

This survey was taken by Institutes of Electrical and Electronics Engineers (IEEE) and Electric Power Research Institute (EPRI). From Fig.1, it is clearly observed that the major occurrences of induction motor faults are on the rotating shaft as well as stator winding side.

A. Stator Faults

Less than 40 % of all reported induction motor failures fall into this category [3]. These winding faults are often caused by insulation failure between two adjacent turns in a coil. The resultant induced current, produces extra heating and causes imbalance in the magnetic field. If undetected, the local heating will cause further damage to the stator insulation until catastrophic failure occurs. The unbalanced magnetic field can also result in excessive vibration and can cause premature bearing failures [2].

The graphical representation of stator faults is shown in Fig. 2 [5]. It shows that all the stator faults occur in the winding.

These faults are called coil-to-coil, turn-to-turn, phase-to-phase, phase-to-ground and open-circuit faults. It has been shown in Fig. 1 from the surveys that the stator fault occurrence has second priority. Therefore, it is important to diagnose these faults at the early stage.

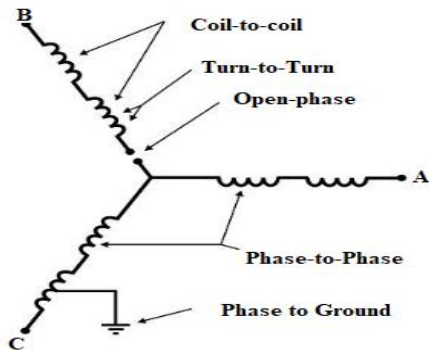


Fig. 2 Graphical Representation of Stator Faults

B. Rotor Faults

Rotor faults occur about almost 10% of total induction motor faults. These faults are caused by rotor windings. They are mainly broken rotor bars because of pulsating load and direct on-line starting. It results into fluctuation of speed, torque pulsation, vibration, overheating, arcing in the rotor and damaged rotor laminations [6]. Fig. 3 shows a broken rotor bar.

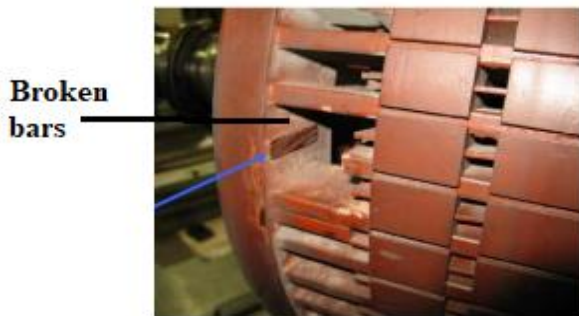
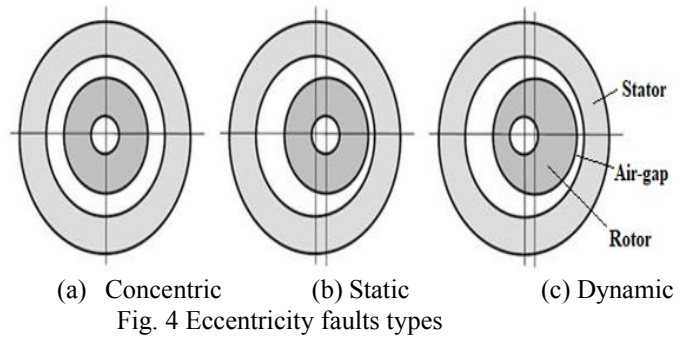


Fig. 3 Broken rotor bar

C. Eccentricity

Air gap eccentricity is due to uneven air gap between rotor and stator of an induction motor. In the case of healthy motor, the rotor is centre aligned with the stator bore. When the rotor is not centre aligned, the unbalanced radial forces cause stator to rotor rub, consequently, damage both the stator and the rotor.

Eccentricity can be categorised into static, dynamic and mixed eccentricity. Mixed eccentricity is a combination of static and dynamic air-gap eccentricity. The minimal radial air-gap length is fixed in space for static eccentricity while the rotor's center and the center of rotation never coincide for dynamic eccentricity [7]. While commissioning of induction motor, a faulty positioning of the stator or rotor may result in static eccentricity. Dynamic eccentricity is due to bearing wear, bent in shaft and mechanical resonances at critical speeds [8]. Fig. 4 shows the types of eccentricity faults.



D. Bearing fault

Usually, a bearing with rolling-element comprises two concentric rings which is a set of rollers spin or balls in the raceways between inner ring and outer ring. Bearing faults may be classified as distributed and local [8]. Distributed faults encompass rough surface, off-size rolling elements, and misaligned races. Localised faults encompass pits, spalls and cracks on the rolling surfaces. When a running roller passes over the fault surface, it generates a streak of vibration impacts at that very instant. The period and amplitude of the impact are calculated by the anomaly's position, speed and bearing dimension [8].

Faults in the inner raceway, outer raceway or rolling elements will produce unique frequency components in the measured machine vibration and other sensor signals. These bearings fault frequencies are functions of the bearing geometry and the running speed. Inner and outer race defects are shown in Fig. 5.

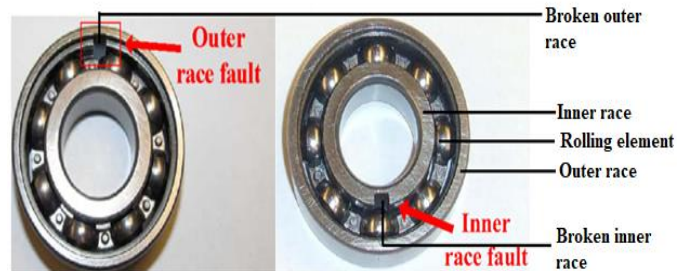


Fig. 5 Outer and inner race bearing defects

III. 3-PHASE INDUCTION MOTOR DESIGN

In this work, a default modeled induction motor in MATLAB/Simulink was used and simulated under various conditions. As a result, all the default parameters were changed to mimic a real industrial motor and these parameters were stored in an m-file. These parameters were accessed by the model while it is running. The parameters include; rated voltage = 400 V, frequency = 50 Hz, speed = 1440 rpm, stator resistance = 0.7384 Ω and rotor resistance = 0.7402 Ω . The stator and rotor self-inductances are equal to 0.003045 H. The mutual inductance between a stator winding and any rotor winding is equal to $L_{sr} = L_m = 0.1241$ H, pole pairs $p = 2$ and the inertial constant $J = 0.000503$ kg.m². These parameters were chosen based on a real motor's parameters. The designed model of the system is shown in Fig. 6.

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} \quad (2)$$

where, I_{max} = RMS value for each stator current and I_{rms} = peak value of the stator current for each phase

These RMS values of the stator current serve as inputs to the fuzzy logic controller. The fuzzy controller assesses these values based on the rules written in its rule base. It uses this information to determine whether a fault has occurred and the severity of this fault.

IV. FUZZY LOGIC CONTROLLER DESIGN

The fuzzy logic theory was introduced by Lotfi Zadeh in 1965 to increase the ability of controllers to cope with problems of uncertainty and imprecision. Fuzzy logic is basically a multivalued logic that allows intermediate values to be defined between conventional evaluations like yes or no and true or false [9]. In fuzzy logic, everything is a matter of degrees. This controller works different from conventional controllers in that, expert knowledge is used instead of differential equations to describe a system [10]. The block diagram of fuzzy logic model is shown in Fig. 7.

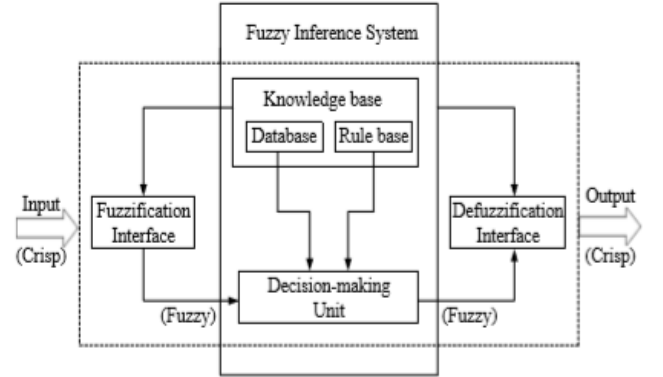


Fig. 7 Block Diagram of Fuzzy Logic Model

Trapezoidal Membership Function (MF) with five partitions were used for the design of the fuzzy logic controller because, it gives better performance in terms of accuracy [11]. In the design, three inputs to the fuzzy controller named I_{as} , I_{bs} and I_{cs} and one output named motor condition were used.

The input and output variables were fuzzified into five trapezoidal membership functions as shown in Fig. 8, where, low, normal, high, very high and too high were the linguistic variables for the input membership function. The membership function for the output in linguistic forms are, open phase, damaged, critically damaged, seriously damaged and healthy.

The range of the input variables I_{as} , I_{bs} and I_{cs} is 0 to 200. The range of the output variable is 0 to 10. The range of the inputs were obtained based on the RMS values of the stator currents produced under various conditions from the simulation of the 3-phase induction motor under steady state and the output variable was chosen arbitrary.

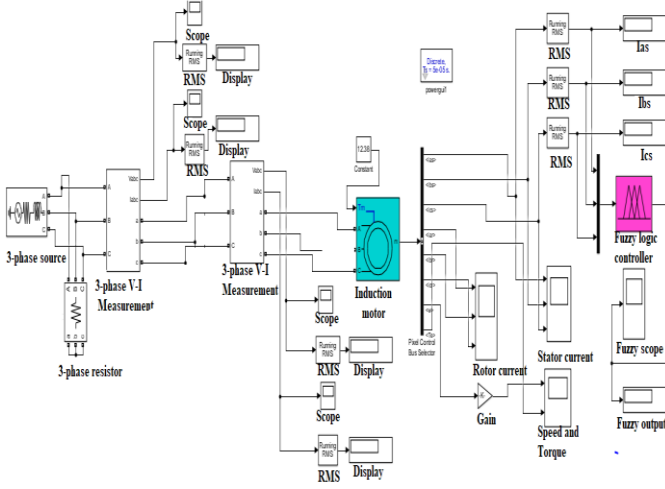


Fig. 6 Simulink model of the designed system

The 3-phase source block is the supply to the induction motor. This source is connected in star configuration with each of the 3-phases having 326.60 V peak value, 50 Hz frequency and displaced 120° from each other. The peak value of each phase is as a result of the motor having a 400 V Root Mean Square Value (RMS) rating. This peak value was computed with an equation expressed as:

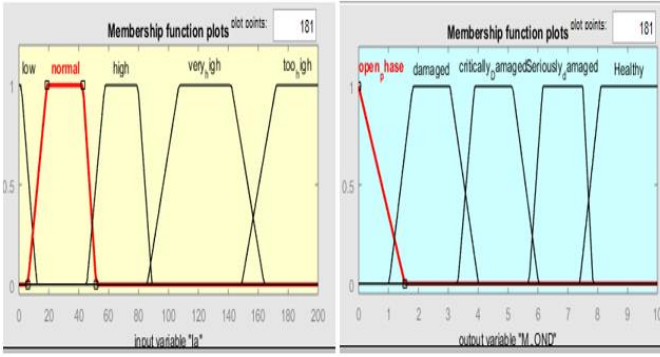
$$V_{max} = \frac{V_{rms} \sqrt{2}}{\sqrt{3}} \quad (1)$$

where, V_{max} = peak voltage value of each phase of the 3-phase source and V_{rms} = root mean square voltage rating of the motor

The supply then passes through a 3-phase V-I measurement block before it is fed to the induction motor. This block basically measures the 3-phase voltage and current of the source. Scopes and display blocks were connected to show these parameters.

When the motor is run, the 3-phase stator current, rotor current, speed and torque generated are displayed in the three scopes labelled stator current, rotor current, speed and torque respectively as shown in Fig. 6. The gain block connected to the speed and torque scope converts the speed of the induction motor from radians per second to revolutions per minute. This is done by multiplying $30/\pi$ by the speed in radians per second.

The 3-phase stator current were then given to the fuzzy logic controller through the RMS blocks shown in Fig. 6. These blocks compute the RMS values of each phase when the motor is under steady state. The equation employed by the RMS block can be expressed as:



(a) Input MF for phase Ia (b) Output MF
Fig. 8 Input and Output Membership Function

The fuzzy inference system used was Mamdani fuzzy inference system because it is the most commonly used inference system in many applications due to its widespread acceptance [12]. The rules for the fuzzy logic controller were written based on the RMS values of stator current amplitudes when the model was simulated.

With the input variables having five memberships functions, the total number of rules for the output variables are $5^2 = 25$ rules. However only nine rules were written based on the faults simulated. The rules are in the If-then format as illustrated below.

1. If Ia is low and Ib is low and Ic is low, then motor condition is open phase.
2. If Ia is normal and Ib is normal and Ic is normal, then motor condition is healthy.
3. If Ia is high and Ib is high and Ic is high, then motor condition is damaged.

The fuzzy rule editor for the input and output variables containing the nine rules is shown in Fig. 9.

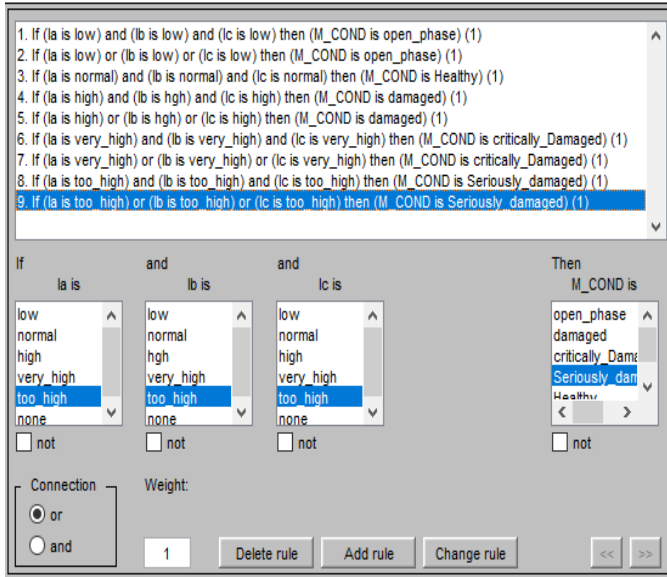


Fig. 9 Fuzzy rule editor

Inference input conditions of stator currents were mapped with consequential output motor conditions, and output on the health of the motor at any instant of time is derived. It is followed by the process of defuzzification for computing crisp indication of motor condition, based on the fuzzy output generated by rule process of the fuzzy inference. There are many types of defuzzification methods available, but the Center of Area (COA) method for defuzzification was used. Despite its complexity it is more popularly used because, if the areas of two or more contributing rules overlap, the overlapping area is counted only once [13].

V. SIMULATION OF INDUCTION MOTOR FAULTS

Simulation of the 3-phase induction motor was also performed by considering various faults conditions like; turn to turn short-circuit in stator windings, break in stator windings, 3-phase to ground and unbalance in input voltage.

A. Turn-to-turn Short circuit in Stator Windings

After the simulation for normal operation of the induction motor model, simulation for the short-circuit in the part of the stator windings was carried out. The stator resistance value of the motor under normal operating condition is 0.7384Ω . The value of the stator resistance at short-circuit fault was assumed to be 0.00542Ω which is far lesser than the normal value. This faulty value means that the resistance has decreased from normal, hence more current will be allowed in the stator windings. The inductance value at the faulty state can be obtained by using the ratio between the values of the resistance at both normal and faulty state. Thus, the value of the inductance was calculated as shown in (3).

$$\frac{R_{\text{stator,normal}}}{R_{\text{stator,fault}}} = \frac{L_{\text{stator,normal}}}{L_{\text{stator,fault}}} \quad (3)$$

$$\frac{0.7384}{0.00542} = \frac{0.003045}{L_{\text{stator,fault}}}$$

$$L_{\text{stator,fault}} = 0.00002235H$$

The stator normal resistance and inductance were substituted with the faulty parameters to obtain the short-circuit response of the induction motor.

B. Break in Stator Windings

Practically, when there is a break in any electrical circuit, it means the impedance at that point has increased to infinity hence permitting no flow of electrical currents. For the purpose of simulation of the break fault in the stator windings, it is not possible to put the value of the stator resistance and the stator inductance to infinity. Because of this, it is assumed that the value of the stator resistance is very large (30000Ω) as compared to value under normal condition and corresponding to this value, the value of the inductance was calculated using (3):

$$\frac{0.7384}{30000} = \frac{0.003045}{L_{\text{stator}, \text{fault}}}$$

$$L_{\text{stator}, \text{fault}} = 123.71$$

The stator resistance and stator self-inductance values were replaced by these faulty values to get the break or open circuited response of the 3-phase induction motor.

C. Unbalanced in Input Voltage

Each of the phases of the supply voltage must be equal in magnitude. However, a fuse in either of the phases may blow or a phase may be short-circuited to ground which may result in unbalance supply to the motor. The simulation of induction motor with voltage unbalance was simulated by simply varying the voltage magnitude in any one of the phases, no other parameter needs to be changed. The fault was created by reducing one of the phases' voltage by 60% and was calculated as:

$$V_{ph3} = V_{ph} - \left(V_{ph} \times \frac{60}{100} \right) \quad (4)$$

$$V_{ph3} = 326.60 - \left(326.60 \times \frac{60}{100} \right)$$

$$V_{ph3} = 145.04 \text{ V}$$

where, V_{ph} = voltage per phase and V_{ph3} = resultant voltage in line three

D. 3-Phase to Ground Fault

Under this fault, all the three phases of the supply voltage were short-circuited to ground. The 3-phase fault block in Fig. 10 was used to introduces this fault. This block provides an option to choose the external fault to be introduced into the system. Example of these faults include, two phase short, two phase to ground and single phase to ground. The fault was introduced into the motor between 0.3 and 1 second of the running period. The Simulink model is shown in Fig. 10.

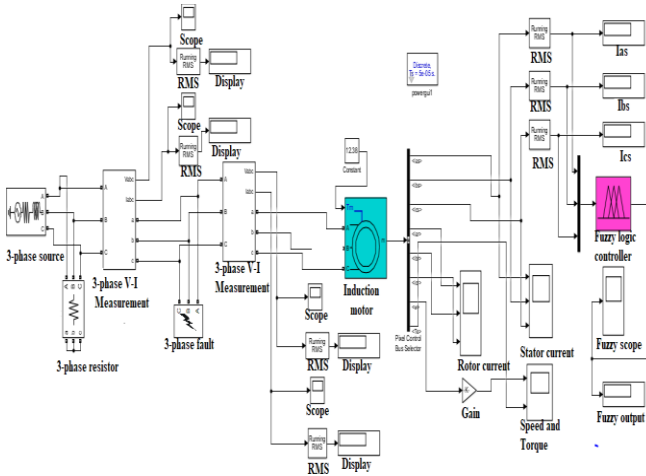


Fig. 10 Simulink model of 3-phase to ground fault

VI. RESULTS AND DISCUSSION

The results obtained when the motor was simulated under various operating conditions are presented and discussed.

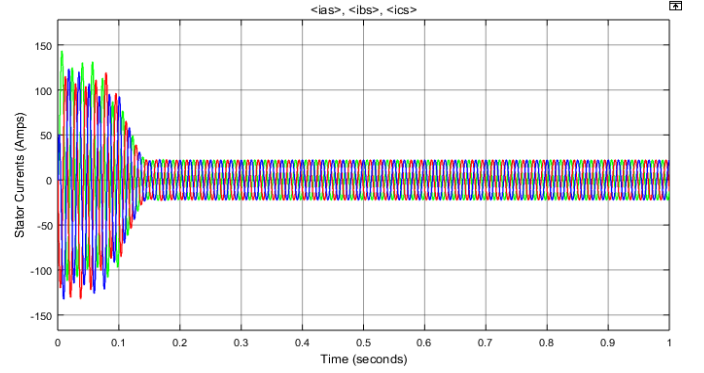


Fig. 11 Stator currents under normal operation

All the phases of the stator current are having same amplitude and are perfectly sinusoidal under steady state. This indicates the healthy condition of the motor.

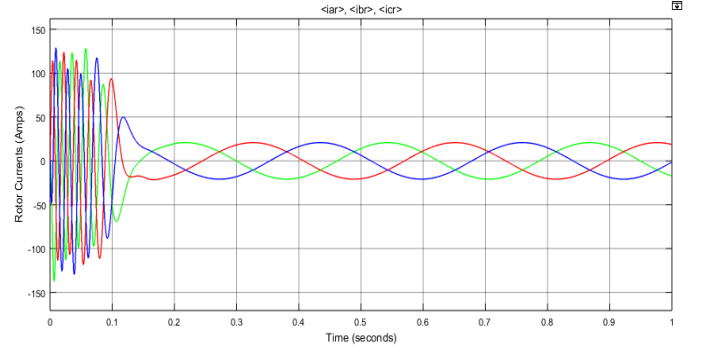


Fig. 12 Rotor currents under normal operation

The rotor currents are also having equal amplitudes under steady state and perfectly sinusoidal, thus, no deviation or any sort of fault has occurred.

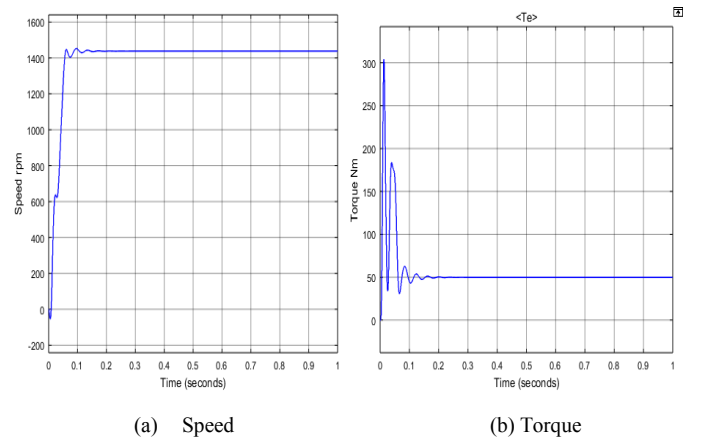


Fig. 13 Speed and torque of the motor under normal operation

The rated speed of the motor was 1440 rpm and as shown in Fig. 13, the exact value was obtained under steady. Moreover, the torque as well did not oscillate as the speed too

did not fluctuate but remained constant throughout the running period under healthy state condition.

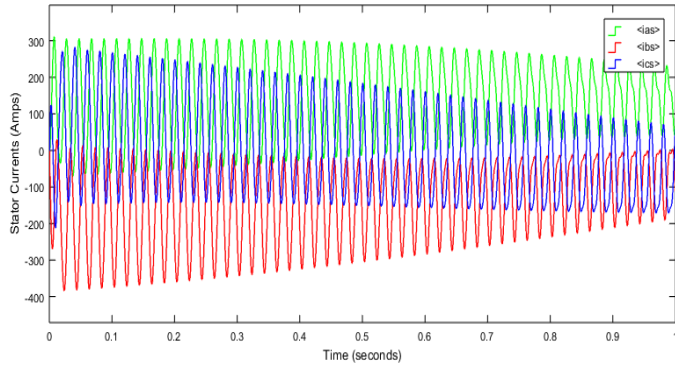


Fig. 14 Stator Current under Turn-to-Turn short-circuit

When this fault occurred, the stator current values increased dangerously high as compared to the normal operation condition as shown in Fig. 14. Comparing this graph to the one in Fig. 11, it can be clearly seen that it has totally deviated from normal. This situation may lead to excessive heating of the motor, arcing, sparks, excessive vibrations, increased noise level and the motor's windings may eventually burn. The red, blue and green colours of the signals represent each of the 3-phase stator current, Ias, Ibs and Ics respectively.

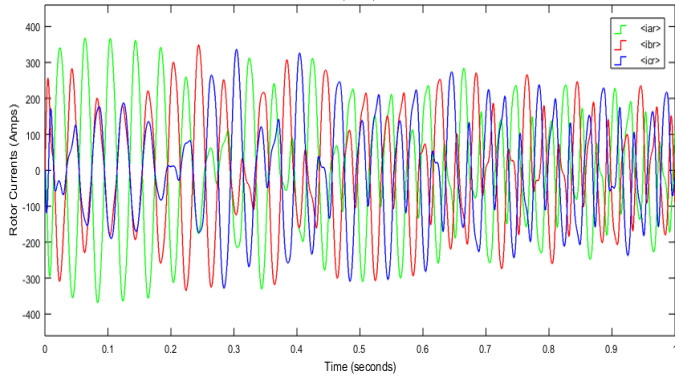
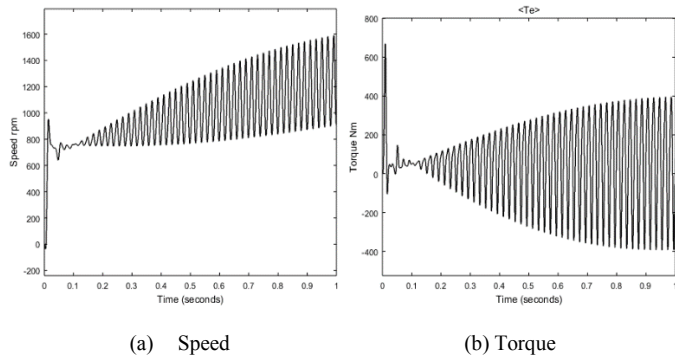


Fig. 15 Rotor Current under Turn-to-Turn short-circuit

The corresponding rotor current values also did increase as shown in Fig. 15. Moreover, it has deviated from being perfectly sinusoidal. The stator and the rotor graph differ as a result of the difference in their frequencies due to slip speed.



(a) Speed (b) Torque
Fig. 16 Speed and Torque under Turn-to-Turn short-circuit

The corresponding torque and speed graph obtained is given in Fig. 16. There is fluctuation in the speed to the point of instability and the torque is not constant but rather it is pulsating. This affect the smooth operation of the motor.

The values of the stator and rotor currents were almost zero (milliamperes) when this condition was simulated. The stator current produced a very minimal field flux. Hence the starting torque generated was not enough to overcome the shaft torque or the load torque. The result for the stator current is shown in Fig. 17.

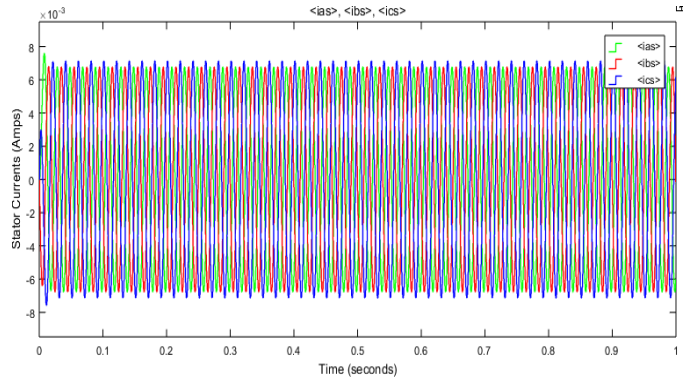
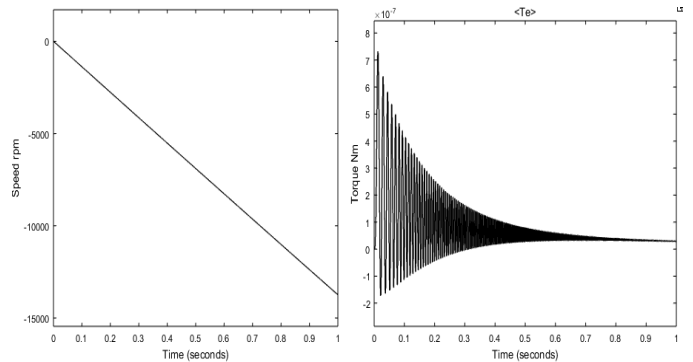


Fig. 17 Stator Current under Break in Stator Winding

The torque oscillates and gradually reduces to zero with the speed going to negative. When this happens, the motor comes to a sudden stop. Fig. 18 shows the speed and torque during this faulty condition.



(a) Speed (b) Torque
Fig. 18 Speed and Torque during Break in stator winding

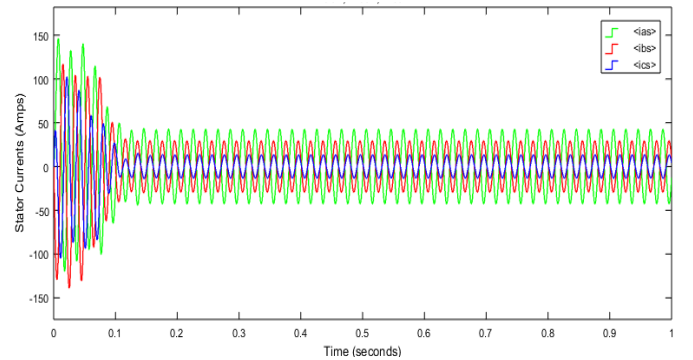
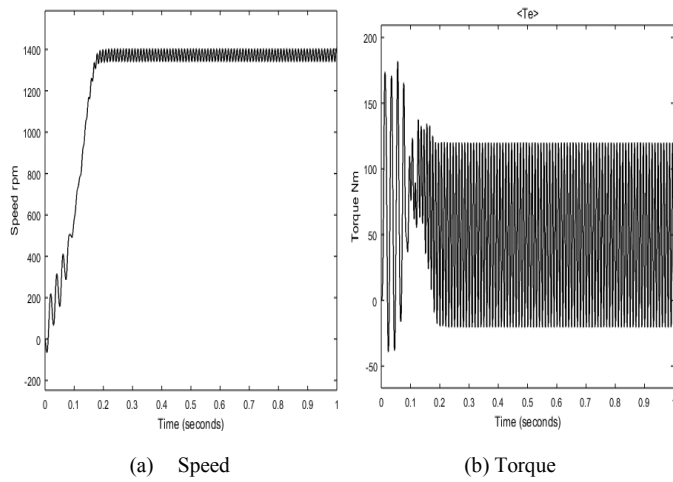


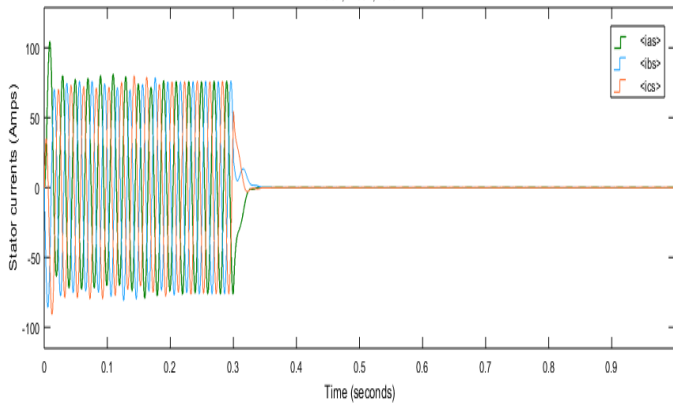
Fig. 19 Stator Current during Unbalance in Input Voltage

The amplitudes of the supply voltage were varied. As shown in Fig. 19, all the stator currents are having different amplitudes. When this occurs, the phase or phases whose amplitude were maintained will draw excess current to help generate the necessary torque required for the motor's operation. This may result in excessive heating of the motor.

The speed of the motor under this condition averagely reduces small and pulsate. The torque also oscillates dangerously due to the imbalance. Fig. 20 shows this torque and speed under unbalanced input voltage.



Under the 3-phase to ground fault, all the three phases of the supply were shorted to ground between 0.3 and 1 second of the running period of the motor. When this occurred, the stator of the motor cease from obtaining current from the supply as depicted in Fig. 21. This will bring the motor to a standstill.



The numerical values of the stator currents obtained during the simulation of the designed system under various conditions and the rule weight for the Fuzzy Logic Controller (FLC) corresponding to the motor's condition are given in Table I. The rule weight is basically the range of values that corresponds to each output MF. Example, between 7.5 to 10 corresponds to healthy, between 0 to 1.5 corresponds to open phase and etc.

The values of the stator currents under each condition has a corresponding rule weight that the FLC uses to conclude the condition of the induction motor based on the rule base. The output results were obtained from the fuzzy scope and the fuzzy output display and they were highly accurate.

TABLE I. SIMULINK RESULTS UNDER VARIOUS CONDITIONS

SN	Mode of Operation	Stator Currents			Fuzzy Logic Controller Output	
		I _{as}	I _{bs}	I _{cs}	Rule Weight	Motor Condition
1.	Normal Operation	24.04	24.04	24.04	Between 7.5 to 10	Healthy
2	Unbalance Input Voltage	46.52	37.28	20.86	Between 1 to 4	Damaged
3.	Break in Stator Winding	0.0047	0.0047	0.0047	Between 0 to 1.5	Open Phase
4.	Turn-to-Turn Short-Circuit	174.60	174.60	174.60	Between 5.5 to 7.5	Seriously Damaged
5.	2-phase Short-Circuit	11.04	25.30	28.34	Between 1 to 4	Damaged
6.	3-Phase to Ground	0.013	0.013	0.013	Between 0 to 1.5	Open Phase

VII. CONCLUSION

The design of the fuzzy logic controller to monitor the condition of the 3-phase induction motor has been achieved. Moreover, from the simulation results, the fuzzy logic controller was able to detect the faults with high accuracy.

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